

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2069

State Street 230 kV
March 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2069. The Interconnection Customer proposes a 111.0 MW solar photovoltaic facility interconnecting at the State Street 230 kV substation in the Indiana area. The request was received on April 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 724 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$34,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SEGS 5 - PEAVINE 100 kV
Contingency: STATE ST SUBSTATION - SEGS 5 100 kV
Loading: 108.6% (Rating: 203 MVA)
DFAX: 7.6%
Pre-project loading: 104.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SEGS 5 - PEAVINE 100 kV	Reconductor line	\$983,000	4% (\$34,000)
	Total allocated to project		\$34,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.